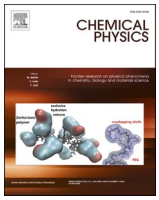
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Dynamical properties of nanosized-water droplets

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ABSTRACT

Molecular dynamics simulations were performed to explore both the bulk and surface regions of the water droplets and slab models. These models were evaluated using the number of hydrogen bonds defined by Wernet et al. and the order parameter tetrahedrality defined by Chau et al. In the bulk region, most of the water molecules formed four saturated hydrogen bonds, while in the surface region, they formed two or three distorted hydrogen bonds. In the surface region of a 400-molecule droplet, water molecules with three hydrogen bonds adopted a tetrahedral configuration as the temperature rose. As the temperature increased, the surface region increased in both models. The diffusion coefficients of the 46-molecule water cluster were larger than those of the bulk droplets, even in the droplet core. These results indicate that all water molecules comprised of the nanosized-droplets were active, as in the surface region.

1. Introduction

Water is the most ubiquitous substance on Earth and has unique properties [1]. For example, its density reaches a maximum at 4 °C [2], it has a high surface tension [3], and the compressibility of liquid water increases under supercooling [4,5]. This is due to the complex network of hydrogen bonds (H-bonds) formed by water molecules. Understanding the reasons for water's remarkable properties is an area of ongoing scientific research.

One of the most important areas of current research in water chemical physics is the existence of a second critical point in deeply supercooled water. Poole et al. proposed that liquid water undergoes a first-order liquid-liquid transition, in which two metastable high-density liquid (HDL) phases and low-density liquid (LDL) phases coexist [6–10]. The existence of this transition is believed to explain the many peculiar properties of liquid water. Experimental proof of the second critical point is difficult to obtain owing to rapid crystallization under supercooled conditions. However, recent experiments have broken through this bottleneck and provide stronger evidence for the existence of the liquid-liquid critical point (LLCP) [11,12]. Theoretically, a metastable liquid-liquid critical point was confirmed by classical molecular dynamics (MD) simulations using TIP4P/2005 and TIP4P/Ice force fields [13].

Nanosized-water droplets have been extensively studied in surface science, nucleation [14], membrane technology [15], and atmospheric chemistry [16,17]. Recent progress in experimental studies on nanosized-water droplets and nanobubbles has been reviewed by Lohse et al. [14]. Bhattacharya et al. [18] studied the nucleation of nanosized-water droplets using in-situ transmission electron microscopy. More recently, Araki et al. [19] observed the behavior of nanosized (<2 nm) -water droplets on a silica glass surface using frequency-modulation atomic force microscopy. For the application of nanosized-water droplets, a humidifier was developed using a conductive polymer material (PEDOT/PSS), and fine water particles with a diameter of ~1.4 nm and no charge were produced. Fine water can improve facial skin moisture levels in adult women [20]. Yang et al. [20] reviewed recent progress in the application of water droplets [21]. Antonov et al. [22] experimentally studied the deformation of water droplets with diameters of 3–6 mm at varying temperatures using a high-speed camera.

To elucidate the unique properties of the aforementioned nanosized-water droplets, theoretical investigations have been performed, particularly MD simulations. Nanosized-water droplets have been applied to various phenomena, such as evaporation kinetics [23,24] and structures on materials [25,26]. Recently, we conducted MD simulations of finite water clusters containing fewer than 400 molecules [27]. In a previous study, the structure of droplets was evaluated using several order parameters, such as tetrahedrality introduced by Chau et al. [28], and g_5, defined by Cuthbertson et al. [29]. However, structure information of

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interfacial water related with hydrogen-bonding motifs is still limited. In the present study, diffusion coefficients were examined to explore the dynamic properties of the water droplets. And temperature dependence of two order parameters examined in Ref. [27] are also reported. For comparison, the droplet properties of the slab model were also analyzed, particularly those of the water-vacuum interface. And we also investigate a finite water cluster consisting of 46 molecules. Due to its nanometer-scale size and the absence of a well-defined bulk region, we refer to this system as a "water nanodroplet" in the sense of a finite-sized cluster, rather than a macroscopic droplet.

2. Methods

We performed all-atom MD simulations using GROMACS version 5.4.1 [30] to investigate the structures of the water nanodroplets. The TIP4P/2005 water model [31] was used, and droplets containing 46, 400, and 1000 water molecules were constructed in the temperature range of 200–360 K. Several water models were proposed and discussed the performance to estimate physical properties such as surface tension [32–35]. According to their studies, the choice of TIP4P/2005 may be good to simulate water surface. Initially the size of the system was minimized. Subsequently, MD simulations were performed in the NVT ensemble (i.e., constant number of particles, N; volume, V; and temperature, T) for 100 ns with a time step of 1 fs. The vibrations of water molecules were also constrained. Short-range interactions were corrected using the Verlet algorithm [36], with a cutoff distance of 0.9 nm. Velocity-rescaling temperature coupling [37] was used to control the temperature. The long-range electrostatic interactions were corrected using the Particle Mesh Ewald (PME) method [38,39] with a grid size of 0.9 nm. The LINear Constraint Solver (LINCS) algorithm [40] was used to constrain all bonds during equilibration. The trajectories were recorded every 10 ps.

The droplet size was measured by the radius of gyration:

$$R_g^2 = \frac{1}{M}\sum_i m_i(\mathbf{r}_i - \mathbf{R})^2 \tag{1}$$

where M is the total mass, $\mathbf{R}$ is the center of mass of the cluster consisting of N molecules, and m_i and $\mathbf{r}_i$ are the mass and coordinates of atom i, respectively.

A bulk water model comprising 1000 water molecules was constructed and energy-minimized before being simulated in the NPT ensemble (i.e., a constant number of particles, N; pressure, P; and temperature, T) for 10 ns with a time step of 0.2 fs. The geometry of the bulk-water model was obtained from a previous study [41]. The isotropic pressure-coupling method was applied at a reference pressure of 1.0 bar using a Berendsen barostat [42]. The short-range interactions, long-range electrostatics, and temperature coupling were identical to those in the NVT ensemble simulations. Both the droplet and bulk water simulations were performed under periodic boundary conditions in all directions. Appropriate boundary conditions were applied to prevent evaporation and to maintain the integrity of the droplets throughout the simulation, even at elevated temperatures.

In the present study, a slab model of liquid water was applied to describe the water-vacuum interface. A simulation cell was first filled with 5000 water molecules and carried out in the NPT ensemble for 3 ns with a time step of 1 fs. Next, the water molecules were placed in a box with the two surfaces oriented perpendicular to the z-axis, allowing for ~5 nm of vacuum on each side of the water layer. An NVT ensemble was then applied for 1 ns with a time step of 1 fs.

The diffusion coefficient, D, is a transport coefficient that can be expressed within the linear response theory as the infinite time integral of an equilibrium time correlation function. In three dimensions it is given by

$$D = \frac{1}{3}\int_0^\infty dt\, \langle \boldsymbol{v}(t)\cdot\boldsymbol{v}(0)\rangle \tag{2}$$

where the variables represent standard quantities. It can also be calculated from the corresponding Einstein relation for slice k:

$$2dtD_k = \left\langle |\boldsymbol{r_k}(t) - \boldsymbol{r_k}(0)|^2 \right\rangle \tag{3}$$

where d is the dimensionality ($d = 1$–3). This relation is valid in the limit that $t \longrightarrow \infty$. The expression in Eq. (3) relates the macroscopic diffusion coefficient to the mean-square distance that the labeled particles travel at time t and is used to calculate the diffusion coefficients in the present case. Because molecular motion at the water surface intrinsically anisotropic, the diffusion coefficient for the slab model was evaluated by decomposing the displacement into components parallel and perpendicular to the instantaneous local surface. We focus on the tangential (in-plane) diffusion, which exhibits a well-defined diffusive regime. For water droplets, the radial and tangential components of the diffusion coefficient were calculated by projecting molecular displacements onto the local direction defined with respect to the droplet center of mass.

The H-bond lifetime is widely used to understand the dynamics of H-bond formation and breakage between water molecules. The H-bond lifetime of water was introduced as proposed by Rapaport [43] and developed by Luzar and Chandler [44] using time correlation functions. The present continuous H-bond correlation function for slice k $S_k(t)$ is estimated

$$S_k(t) = \frac{\langle h_k(t_0)\cdot H_k(t_0, t_0+t)\rangle_{t_0}}{\langle h_k\rangle} \tag{4}$$

where the variable $h_k(t_0)$ is unity if a given pair of molecules located in slice k is H-bonded at time t_0, and zero otherwise. The function $H_k(t_0, t_0+t)$ is unity if the same pair remains continuously H-bonded throughout the time interval $[t_0, t_0+t]$, and zero otherwise. The brackets $\langle\cdots\rangle_{t_0}$ denote averaging over all possible time origins t_0 and over all H-bonded pairs. The obtained correlation functions exhibit a clear two-step decay and were fitted using a bi-exponential function,

$$S_k(t) = A_k \exp\left(-\frac{t}{\tau_{1,k}}\right) + (1-A_k)\exp\left(-\frac{t}{\tau_{2,k}}\right) \tag{5}$$

where $\tau_{1,k}$ and $\tau_{2,k}$ correspond to the fast (librational) and slow (structural relaxation) components, respectively. The average H-bond lifetime was evaluated as the time integral of the fitted correlation function,

$$\tau_{HB,k} = \int_0^\infty S_k(t)dt = A_k\tau_{1,k} + (1-A_k)\tau_{2,k} \tag{6}$$

Representive correlation functions together with the corresponding bi-exponential fits are shown in Supplementary data.

In this study, the local structures of the water nanodroplet models, bulk water (1000 molecules, T = 300 K), and slabs were examined in detail. The tetrahedrality of the local structure factor [28] is also examined. The tetrahedrality, Q_i, of the ith water molecule [28] is given by Eq. (5):

$$Q_i = 1 - \frac{3}{8}\sum_{j=1}^{3}\sum_{k=j+1}^{4}\left(\cos\theta_{jik} + \frac{1}{3}\right)^2 \tag{7}$$

where θ_{jik} is the angle formed by the lines joining the jth and kth oxygen atoms with the central (ith) oxygen atom. For a perfect tetrahedral arrangement of water molecules around the central ith oxygen atom, $\cos\theta_{jik} = -1/3$, thus $Q = 1$. For the random arrangement, $Q = 0$. By calculating tetrahedrality, we can approximate the atomic arrangement around the ith oxygen atom.

Another local structural parameter, g_5, [29], defined as the distance between the ith oxygen atom and its fifth-nearest neighbor, was also applied. In addition, g_5 indicates the degree to which additional neighbors disrupt the tetrahedral structure of the first coordination. For a perfect tetrahedral arrangement, g_5 is 4.51 Å.

3. Results and discussion

3.1. Diffusion coefficients of the water liquid-vapor interface and the droplet model

The slab model was delimited along the z-axis, and the measured water-density distribution is shown in Fig. 1(a). The region with a density of 1 g cm^{-3} represents the bulk region, whereas the region with a density of 0 g cm^{-3} represents the region with no water molecules (vacuum). The density fluctuates around 1 g cm^{-3}, and these fluctuations decrease with increasing temperature. This indicates that the water density in the bulk region became uniform with increasing temperature. The width of this abrupt density change (surface thickness) was measured to be 3.2 Å at 280 K and 4.7 Å at 360 K. Previous studies [45,46] have reported that the minimum distance from the aqueous liquid-vapor interface at which bulk liquid behavior is observed is 5–10 Å.

The radii of gyration of water nanodroplets with 46, 400, and 1000 water molecules are summarized in Table 1. These values are consistent with previous estimates of the approximate radii of the nanodroplets for 46 and 400 water molecules [27]. The radius of gyration corresponds to the value at which the water density began to decrease. The region outside the radius of gyration is the liquid-vapor interface, as discussed in the slab model.

The diffusion coefficients delimited by the z-coordinate every 5 Å are shown in Fig. 1(b). A value of 5 Å is larger than the typical oxygen-oxygen distance of 2.95 Å for liquid water and may be suitable as a width that includes two waters. These values were symmetrized by the absolute values of the z-coordinate. In the middle region of the slab model, the diffusion coefficients were comparable to those of bulk water at all temperatures examined in this study. In the surface region, the diffusion coefficient began to increase as the water density decreased.

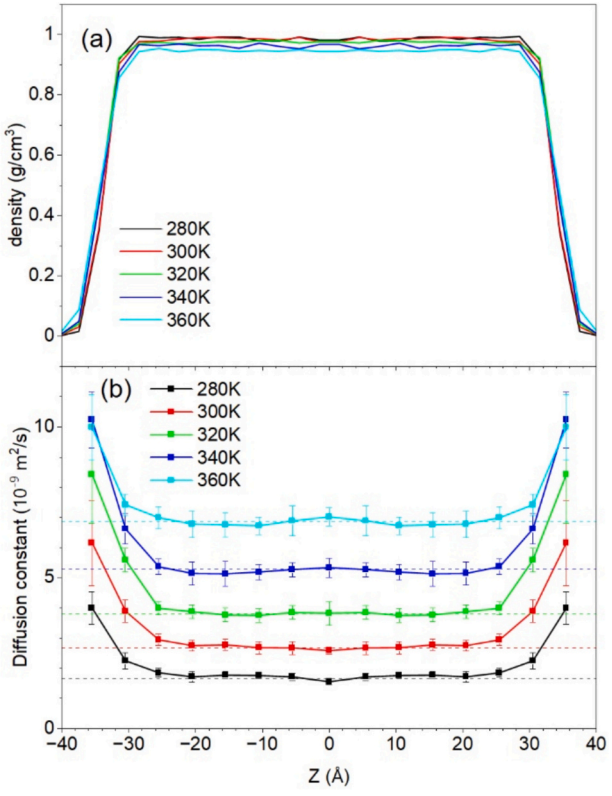


Fig. 1. (a) Density along the z-axis from the center of the slab model. (b) Diffusion constants of as a function of the z-axis. Dashed lines indicate the calculated bulk water. Note that the diffusion constants are symmetrized by those for the same absolute values of the z-axis.

Table 1
Radius of gyration (units in Å) at each temperature of the model droplets including 46, 400, and 1000 molecules.

Temp.	280	300	320	340	360
46 mol. droplet	5.44	5.52	5.57	5.65	5.90
400 mol. droplet	11.10	11.16	11.26	11.36	11.49
1000 mol. droplet	15.01	15.11	15.26	15.50	15.89

This can be attributed to the weakness of H-bonds in the surface region.

The H-bond dynamics in the air-water interface using the slab model have been extensively studied by Liu et al. [47]. The H-bond dynamics near interfaces exhibit different kinetics than in bulk water. MD studies have shown that both H-bond breaking/reformation rates and lifetime correlation functions depend on molecular environment and distance from the interface.

The diffusion coefficients of the 400 water molecules in a droplet as a function of temperature are shown in Fig. 2 (a). For reference, the values for bulk water are shown in the same figure; the coefficients increase with increasing temperature. In the region less than 10 Å from the center of the droplet, the estimated diffusion coefficients were compared with those of the bulk. This value is consistent with the gyration of a 400-water molecule droplet. For regions larger than 10 Å, the diffusion coefficients became larger than those of the bulk at all temperatures. Thus, the droplet comprising 400 water molecules included both the bulk and the surface. Fig. 2 (b) shows the size dependence of the diffusion coefficients of water molecules as a function of the radius from the center of the droplet at 300 K. Diffusion coefficients less than the radius (15 Å) are compatible with the bulk for 1000-water-molecule droplets. Thus, analogously to the case of 400 water molecules, these droplets include both the bulk and surface regions. However, for the 46-water-molecule droplet, even at the smallest radius (2.5 Å), the diffusion coefficient was larger than that of the bulk. In this study, the surface thickness is smaller than this distance: the radius of a 46-molecule droplet is ~5 Å [27], and considering the above results, the radius of the bulk region is ~2 Å, which is sufficiently large for a single water molecule to exist. Thus, a small droplet has almost no bulk region, and similar to the surface, all the water molecules are active. Using the thickness of the surface, we divided the 400-molecule droplet into bulk and surface regions, and performed several analyses. In these figures, radial and tangential components are displayed separately. For bulk regions, both components are compatible, on the other hand, for surface regions, the radial component is significantly suppressed due to geometric confinement. The radial motion does not reach a fully diffusive regime within the analyzed time window.

The H-bond lifetime of the 400 water molecules in a droplet as a function of temperature are shown in Fig. 3. In these figures, the average H-bond lifetimes τ_{HB} in Eq. (6) are shown together with the fast and slow relaxation components τ_1 and τ_2. As increasing temperature, the lifetime decreases. In the bulk regions, these are compatible at all temperatures. At lower temperatures, the surface region (10–15 Å) exhibits a longer slow relaxation component (τ_2) than the interior. Although surface molecules have fewer hydrogen-bond partners, their reorientation and exchange dynamics are partially suppressed due to reduced coordination and geometrical constraints. This results in a slower network relaxation process at the surface. The crossover around 340 K originates from enhanced thermal fluctuations that significantly increase surface mobility. Above this temperature, surface molecules undergo faster reorientation and hydrogen-bond switching than interior molecules, leading to a shorter surface lifetime.

Fig. 3 (b) shows the size dependence of the H-bond lifetimes as a function of the radius from the center of the droplet at 300 K. The non-monotonic size dependence arises from competition between surface-dominated constraints and bulk-like cooperative rearrangements. The 46-molecule droplet lacks a fully developed hydrogen-bond network, resulting in shorter lifetimes. The 400-molecule droplet exhibits optimal

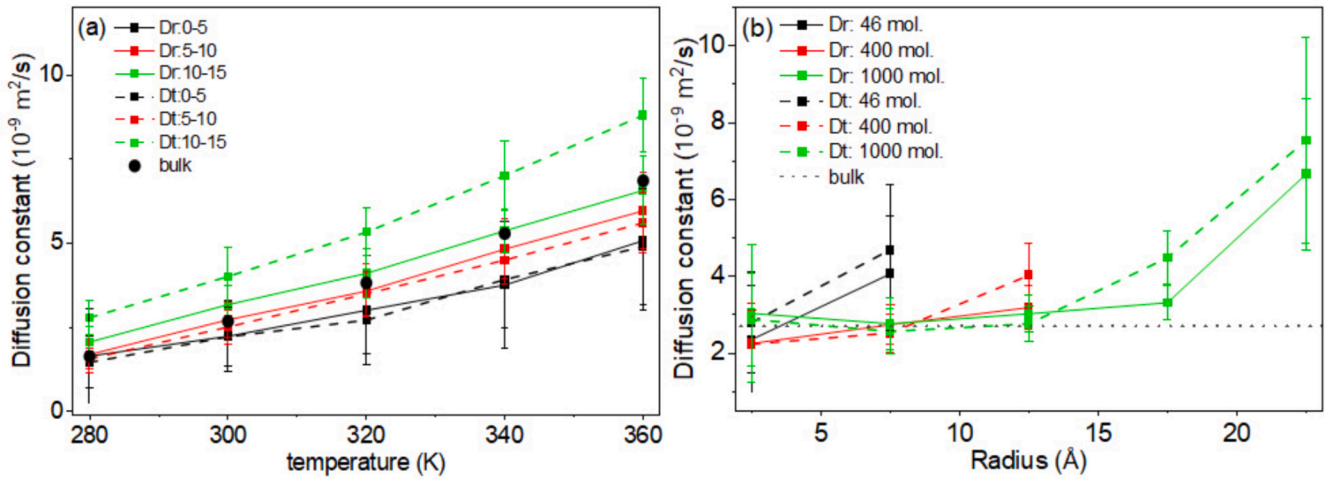


Fig. 2. (a) Diffusion constants in 400-molecular droplets as a function of temperature. The corresponding bulk water is represented by black circles. Squares with solid and dashed lines correspond to the radial (D_r) and tangential (D_t) components respectively. (b) Diffusion constants for 46, 400, and 1000-molecular droplets at 300 K as a function of droplet radius. The corresponding bulk water is represented by black dotted line.

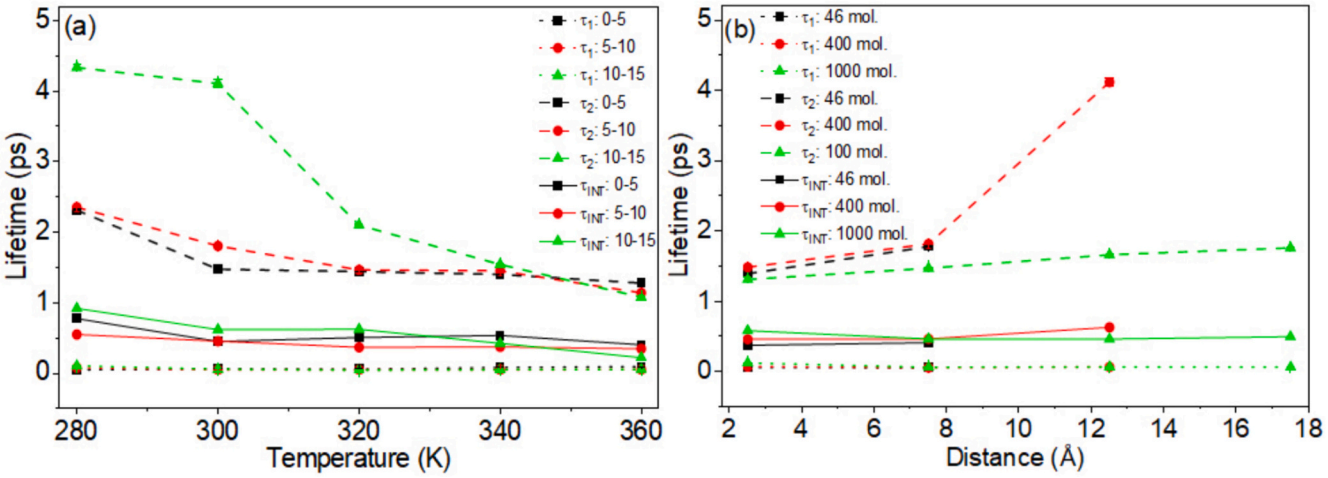


Fig. 3. (a) H-bond lifetimes as a function of temperature. (b) H-bond lifetimes at 300 K for 46, 400, and 1000-molecular droplets as a function of droplet radius.

network stabilization. In contrast, the 1000-molecule droplet shows more bulk-like collective rearrangements that facilitate hydrogen-bond exchange, slightly reducing the lifetime compared to the 400-molecule case.

3.2. Distribution of Q and g_5 grouped by the number of H-bonds

To gain a deeper understanding of the distribution of Q from a previous study [27], we classified Q into different types of H-bonds. We

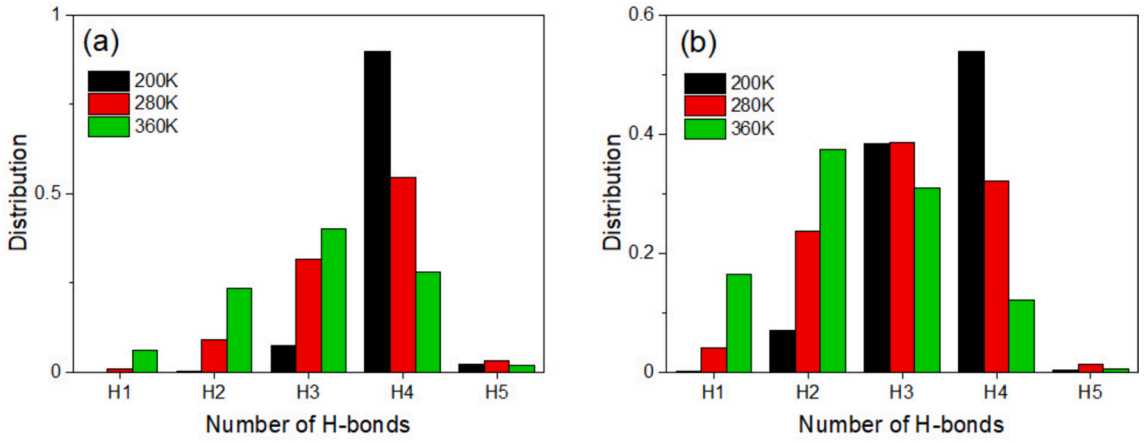


Fig. 4. Distribution of the number of H-bonds in 400-molecules droplets at 200, 280, and 360 K. (a) Bulk and (b) surface regions.

applied the H-bonding criteria proposed by Wernet et al. [48] to examine the formation of H-bonds, which is based on the combined constraints on the intermolecular O—O distance and the H-bond angle. More explicitly, the following function $r(\theta)$ is used

$$r(\theta) = -0.00044\,\theta^2 + r_{max} \tag{8}$$

where θ is O…O-H angle (in degree) and r_{max} (in Å) is a threshold value, and we applied as 3.3 Å, which is obtained to conventional criteria typically applied to MD simulations. In the present study, the droplet and slab models were classified by the number of H-bonds, or Hm, where m is the sum of the number of H-bond donors and acceptors, and the H-bond donors and acceptors were not distinguished.

Fig. 4 shows the distribution of the number of H-bonds in the 400-molecule droplets obtained by analyzing the geometry of the droplets in the MD simulation. The distributions in Fig. 4 (a) and (b) show the bulk and surface regions, respectively. H4 and H2 correspond to the tetrahedrally arranged and distorted clusters of water molecules, respectively. As shown in Fig. 4 (a), in the bulk region at 200 K, most of the water molecules formed four H-bonds, whereas in the surface region shown in Fig. 4 (b), the distribution of the three H-bonds increased, indicating that the water molecules in the surface region were more active than those in the bulk. With increasing temperature, the percentage of water molecules forming four H-bonds decreased, whereas those forming one or two H-bonds increased. This means that the molecular motion becomes more aggressive with increasing temperature.

Fig. 5 shows the distribution of Q for different numbers of H-bonds in the 400-molecule droplets. In the bulk region, H4 was dominant at 200 K, and by increasing the temperature, more distorted configurations, such as H2 and H3 became more prevalent, as shown in Fig. 4. Furthermore, the Q peak for H4 decreased slightly with increasing temperature; specifically, it changed from 0.925 at 200 K to 0.825 at 280 K and 0.725 at 360 K. For H2 and H3, only trace amounts were observed at 200 K and were distributed over a wide range at 280 K. Distinct peaks were observed at 0.475 for H2 and 0.525 for H3 at 360 K, which were smaller than those of H4. This indicates that a water molecule forms a regular tetrahedron surrounded by four other water molecules; however, their arrangement gradually distorts with increasing temperature, even in the H4 configuration. In the surface regions, compared with the bulk, the distribution of distorted configurations increased at all temperatures. The H4 peak decreased slightly with increasing temperature, which is the same trend as that for the bulk. The H3 peak shifted slightly with increasing Q.

Fig. 6 shows the distribution of g_5 as a function of the oxygen-oxygen distance, r (in Å) for different numbers of H-bonds in 400-molecule droplets. In all cases, H5 was distributed around 3 Å, which is a typical H-bond distance. In the bulk regions of Fig. 6 (a), 6 (b), and (c), H4 is dominant at 200 K, and by increasing the temperature, more distorted configurations, such as H2 and H3 increase, as shown in Fig. 4. The g_5 peaks for H2, H3, and H4 are stable between 3.25 and 3.35 Å at 280 and 360 K. For H4, all the peaks shifted slightly toward decreasing g_5 with increasing temperature, that is, from 3.65 Å at 200 K to 3.35 Å at 280 K and 3.25 Å at 360 K. In the surface regions of Fig. 6 (d), 6 (e), and (f), the distribution of the surface was slightly wider than that of the bulk. H4 exhibited the same trend as the bulk, and the peak for the surface shifts was larger than that for the bulk at all temperatures. According to Cuthbertson et al. [29], the threshold value between the HDL and LDL phases was 3.5 Å. From this perspective, the droplet consisting of 400 water molecules belonged to the HDL phase. This corresponds to a higher percentage of water molecules existing in the HDL phase. The computational result that small droplets have a peak slightly shifted toward higher r is most likely due to the disordered and largely fluctuating surface structure, not the high tetrahedrality.

Fig. 7 shows the distribution of the number of H-bonds in the slab model. The distributions in Figs. 7 (a) and 7 (b) represent the bulk and surface regions, respectively. In the bulk region, the distribution with respect to the number of H-bonds showed the same trend as in the case of the droplets: at 200 K, H4 was the most distributed in the bulk and H3 on the surface. With increasing temperature, the number of broken H-bonds, such as H2 and H3, increased, and that of H4 decreased. The H2 and H3 in the surface region are higher than those in the bulk..

Fig. 8 shows the distribution of Q in the slab model. In the bulk region, H4 was dominant at 200 K, and by increasing the temperature, more distorted configurations such as H2 and H3 increased, as shown in Fig. 7. Furthermore, the Q peak for H4 decreases slightly with increasing

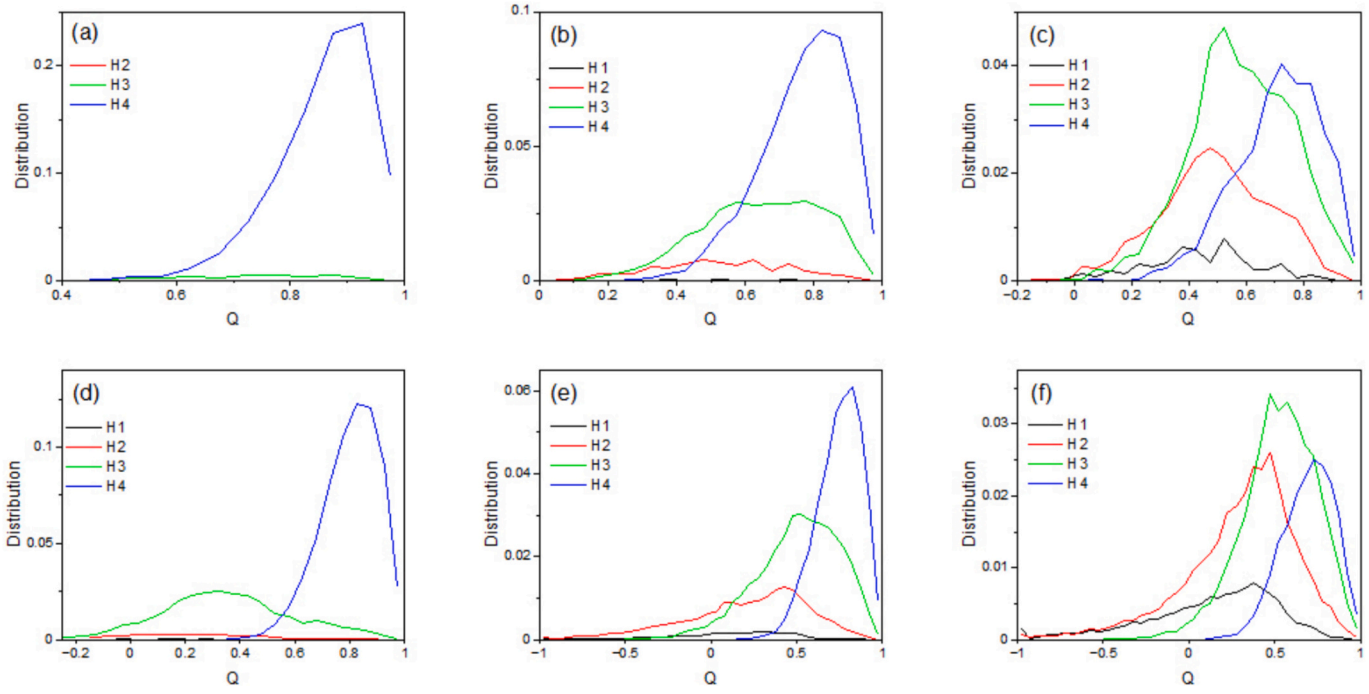


Fig. 5. Distribution of Q with respect to the number of H-bonds in 400-molecules droplets. (a)–(c) Distributions in the bulk region at 200, 280, and 360 K, respectively. (d)–(f) Distribution in the surface region at 200, 280, and 360 K, respectively.

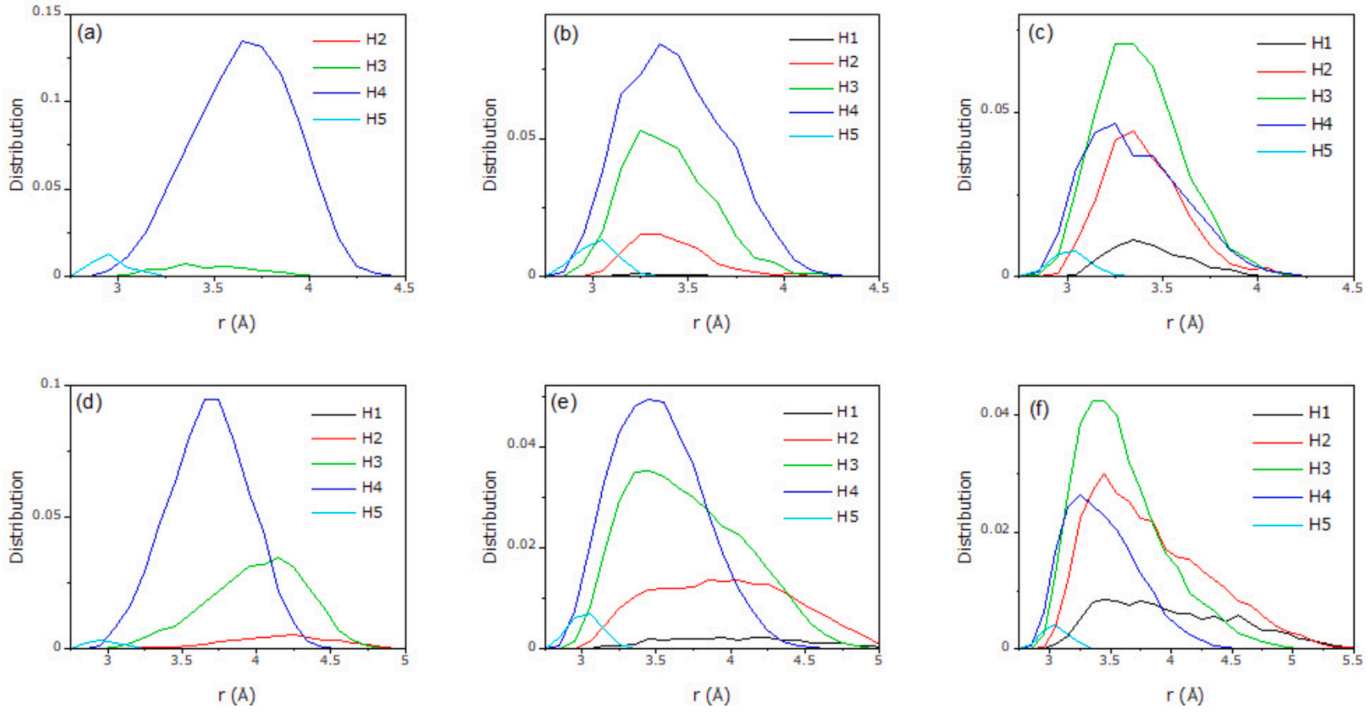


Fig. 6. Distribution of g_5 with respect to the number of H-bonds in 400-molecules droplets. (a)–(c) Distributions in the bulk region at 200, 280, and 360 K, respectively. (d)–(f) Distribution in the surface region at 200, 280, and 360 K, respectively.

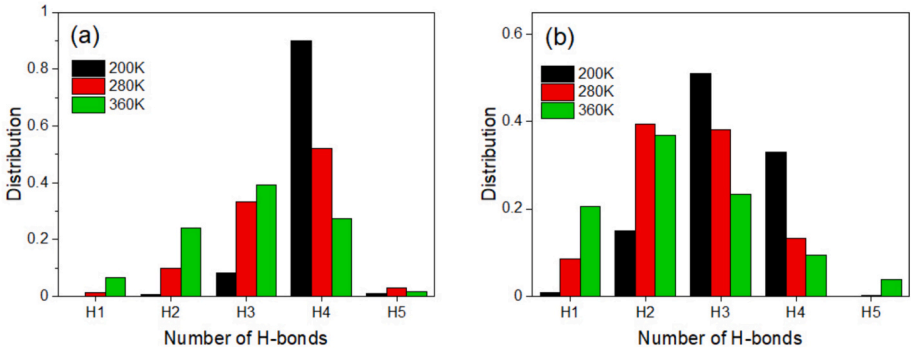


Fig. 7. Distribution of the number of H-bonds in the slab model at $T = 200$, 280, and 360 K. (a) Bulk and (b) surface regions.

temperature. Only trace amounts of H2 and H3 were observed at 200 K, and were distributed over a wide range at 280 K. A distinct peak appears at 0.45 for H2 (0.5) H3 at 360 K, which is smaller than that for H4. In the surface regions, compared with the bulk, the distribution of distorted configurations increased at all temperatures. The peak for H4 decreases slightly with increasing temperature, which is the same trend as that of the bulk. The H3 peak shifts slightly with increasing Q and temperature.

Fig. 9 shows the distribution of g_5 in the slab model. In the bulk regions of Fig. 9 (a), (b), and (c), H4 is dominant at 200 K, and by increasing the temperature, more distorted configurations, such as H2 and H3, increase, as shown in Fig. 6. Peaks of g_5 for H2, H3, and H4 are stable between 3.2 and 3.4 Å at 280 and 360 K, respectively. For H4, all the peaks shifted slightly toward decreasing g_5 with increasing temperature, that is, from 3.8 Å at 200 K to 3.4 Å 280 K and 3.2 Å at 360 K. In the surface regions of Fig. 9 (d), (e), and (f), the distribution of the surface was slightly wider than that of the bulk. H4 exhibited the same trend as the bulk, and the peak for the surface shifts was larger than that for the bulk at all temperatures.

Based on the above results, we examined the growth process on the water surface. At low temperatures, water molecules are randomly added to the surface. Consequently, the Q value for a distorted hydrogen-bond arrangement is small. As the temperature increases and molecular motion intensifies, water with larger Q values is formed.

4. Conclusion

In this study, MD simulations were used to investigate the structure of water molecules in nanometer-sized droplets and slabs, with a particular focus on both the surface and bulk regions. The boundary between the bulk and the surface was determined using the density profile of the droplet, as calculated in a previous study. The model was analyzed using two order parameters: tetrahedrality and g_5.

Density distribution analysis of the slab model revealed that the surface thicknesses were 3.2 and 4.7 Å at 280 and 360 K, respectively. This thickness was used to separate the bulk and surface regions of the droplets for subsequent analyses. The distribution of the droplet and slab models according to the number of H-bonds of water molecules was examined. We found that the density distribution became uniform in the bulk region as the temperature increased, and the tetrahedral structure gradually distorted. The density distribution fluctuated more in the

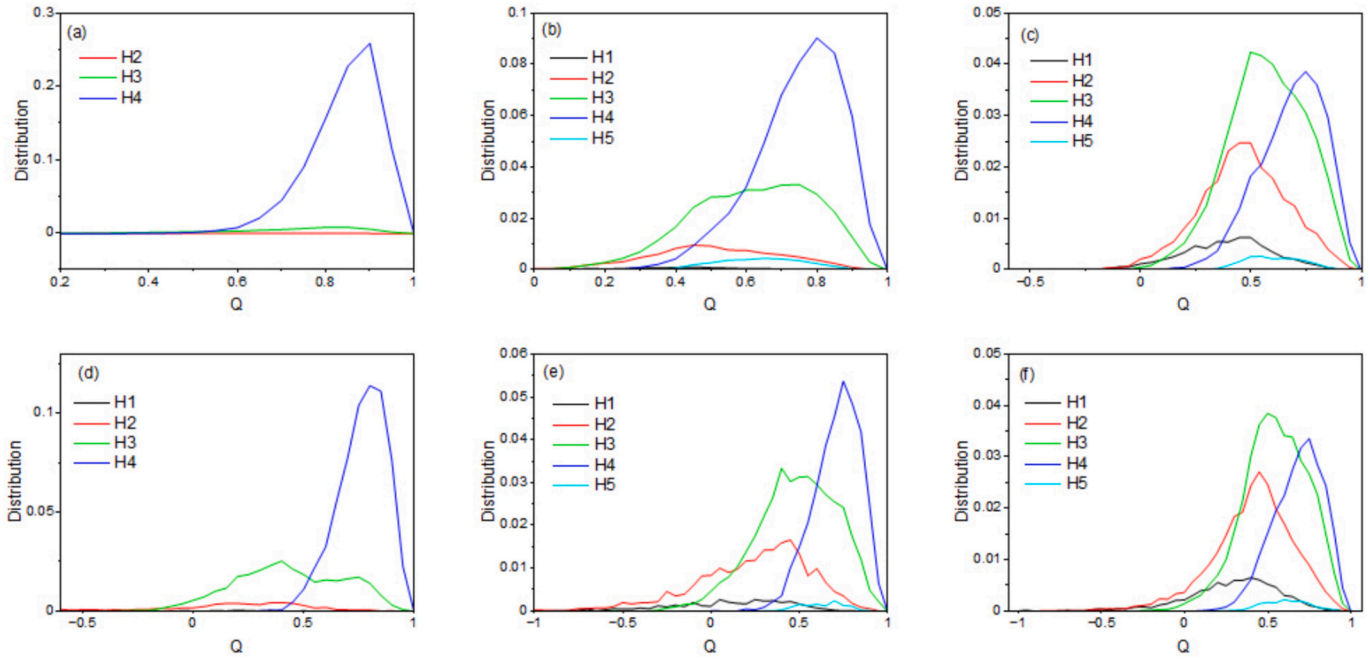


Fig. 8. Distribution of Q with respect to the number of H-bonds in the slab model. (a)–(c) Distributions in the bulk region at 200, 280, and 360 K, respectively. (d)–(f) Distribution in the surface region at 200, 280, and 360 K, respectively.

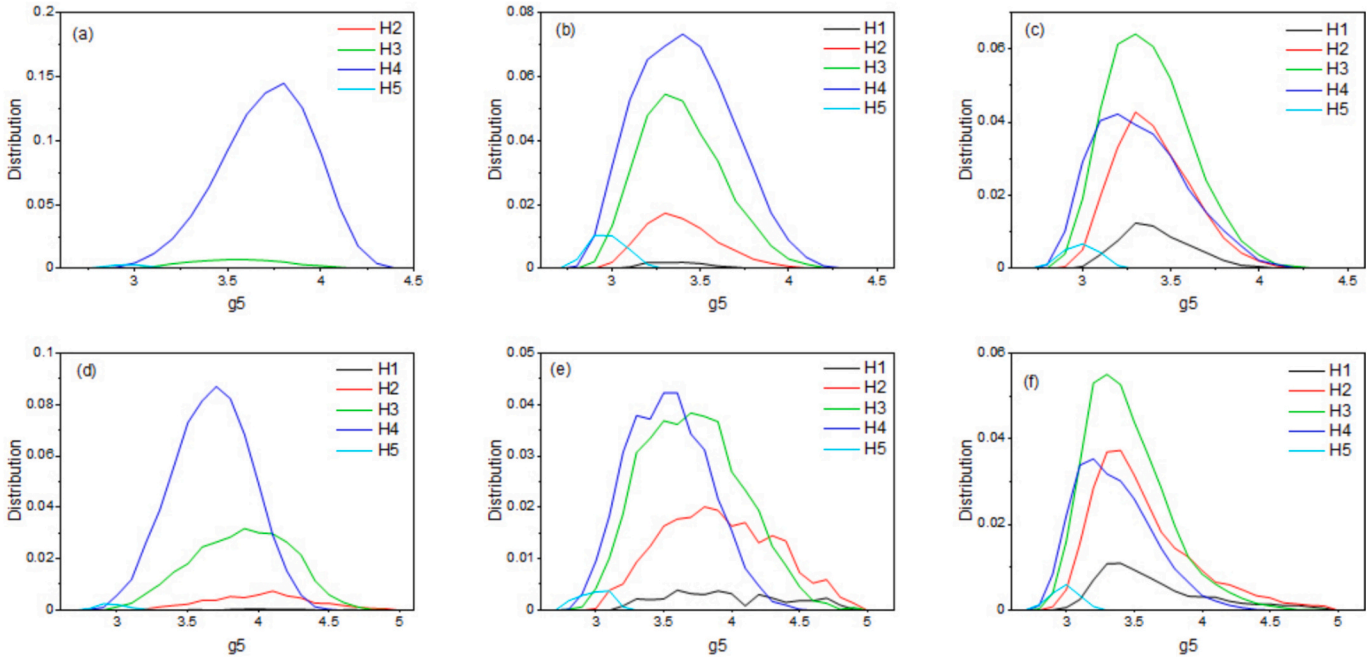


Fig. 9. Distribution of g_5 with respect to the number of H-bonds in the slab model. (a)–(c) Distributions in the bulk region at 200, 280, and 360 K, respectively. (d)–(f) Distribution in the surface region at 200, 280, and 360 K, respectively.

surface region than in the bulk region.

The diffusion coefficients of liquid water for the slab and droplet models were examined as dynamic properties. The calculated diffusion coefficients at each temperature are consistent with the experimental results. The diffusion coefficients of the slab and droplet models were examined, and a good correlation with density was obtained. Thus, diffusion coefficients are good indicators for distinguishing the surface from the bulk. For 46 molecules, the region at the center of the droplet diffusion coefficient was larger than that of the bulk. The 46-molecule water nanodroplet is so small that it does not have a bulk region. Thus, water nanodroplets of this size are more active than those larger than ~10 Å. Our findings are useful for understanding the reactivity and applications of small droplets.

CRediT authorship contribution statement

Nanami Shima: Writing – original draft, Investigation, Data curation. **Osamu Takahashi:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no competing financial interests or personal relationships that may have influenced the work reported in this study.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chemphys.2026.113166.

Data availability

Data will be made available on request.

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